Kallem Maanya Sri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.04
Narayana Junior College	Class XII (Intermediate)	2024	98.80%
Narayana Concept HS	Class X (SSC)	2022	10.00%

SKILLS

Programming Languages: Python, C

Tools & Technologies: LaTeX, Git, GitHub, MySQL

Competitive Programming: Solved 300+ problems on LeetCode | **View Profile here**

Core Concepts: Machine Learning, Deep Learning, Data Structures & Algorithms, DBMS

Soft Skills: Effective Communication, Collaborative Teamwork, Leadership, Initiative

PROJECTS

Custom ML Library & Autograd Engine | *Python, NumPy*

- Designed and implemented a modular software library using **object-oriented design principles**, with a dynamic autograd engine supporting broadcasting and automatic gradient computation
- Built reusable, extensible **APIs** for model construction and training, enabling clean integration across components
- Developed **19 reusable, well-tested components** (optimizers, loss functions, activations, classical ML algorithms) following consistent **design patterns** for maintainability and code reuse

Sequence Models from Scratch | *Python, NumPy, PyTorch*

- Engineered modular training pipelines for RNN, LSTM, Transformer, and State Space Model architectures, emphasizing clean, maintainable, and reusable code structure
- Debugged and quantified the vanishing gradient problem, identifying **~100× gradient decay** in vanilla RNNs vs. stable LSTMs through systematic analysis and troubleshooting
- Benchmarked inference performance across architectures, achieving **74.8% accuracy** on long-context sequence tasks

Deep Learning Models from Scratch | *Python, NumPy, Matplotlib, DL*

- Designed configurable, reusable ANN and CNN architectures from scratch with custom forward/backward propagation
- Debugged and tuned **7 optimizer/regularization configurations** (SGD, RMSProp, Adam, Dropout) to fix bottlenecks
- Improved test accuracy from **10% to 41%** through systematic optimization and debugging

Explainable AI & Adversarial Robustness *(Group Project)* | *Python, PyTorch, LaTeX, Research Engineering*

- Reproduced adversarial attacks on feature attribution methods across ImageNet, CIFAR-10 to evaluate robustness, as part of a 4-person team.
- Quantified adversarial perturbation impact, reducing explanation overlap to 9-16% on ImageNet.
- Compared attribution methods under adversarial settings; identified Integrated Gradients as most robust.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications, Algorithms, DBMS

Artificial Intelligence: Foundations of Machine Learning, Deep Learning, Introduction to Modern AI, Intro to Classical AI, Programming for AI, Optimization I, Optimization II, AI for Humanity

Mathematics: Probability and Random Variables, Statistics, Linear Algebra (Matrix Theory), Calculus, Discrete Mathematics, Numerical Analysis

Others: Linear Systems & Signal Processing, Control Systems

EXTRACURRICULAR

- Event Management: Served as Operations Core in EBSB, contributing to the planning and execution of campus events.
- NSS Volunteer: Participated in community service and social outreach initiatives .

POSITIONS OF RESPONSIBILITY

Internship Coordinator, Office of Career Services (OCS) | 2026 - Present

- Coordinated internship drives, engaging companies and supporting students through the recruitment process.

Sunshine Buddy, Mental Health Club | 2024 - 2025

- Organized icebreaker and wellness sessions to promote student well-being and foster campus inclusivity.

Head, Shuffle Club (Dance Club) | 2026 - Present

- Managing club activities, coordinating events, and overseeing team operations.